



Minor project 101007491 CyberInsight AI

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Introduction

- Cybercrime is rapidly increasing due to widespread smartphone and high-speed internet adoption.
- Users frequently receive malicious APK files, executable files, phishing links, and scam QR codes.
- These threats are commonly distributed through:
 - WhatsApp
 - Telegram
 - Email
 - Social Media
- Users may unknowingly install malicious applications or access harmful links.

- This can result in:
 - Device compromise
 - Sensitive data theft
 - Identity fraud
 - Financial losses

CyberInsight AI is an intelligent cyber-threat analysis platform designed to help users understand and identify potential cyber threats.

Background and Motivation

- Everyday users often cannot determine whether a downloaded file or application is safe before installation or execution.
- Existing cybersecurity tools generally provide complex technical and forensic reports.
- These reports require professional cybersecurity knowledge to understand.
- There is a need for a **beginner-friendly cybersecurity platform**.
- CyberInsight AI bridges this gap by converting complex threat indicators into:
 - Simple explanations
 - Risk scores
 - Actionable recommendations

Major Threats : -

- Fake Banking APKs
- Fake e-Challan APKs
- Trojanized traffic applications
- Malicious EXE files
- Phishing websites
- Scam QR codes

Relevance and Importance

- **Relevance and Importance**

- Cyber threats are becoming increasingly common for ordinary internet users.
- Non-technical users face difficulty interpreting cybersecurity reports.
- CyberInsight AI provides a simple and understandable security analysis platform.
- The platform analyzes multiple threat formats:
 - APK files
 - EXE files
 - Web URLs
- AI converts technical security findings into **plain-language explanations**.
- Users receive direct recommendations such as:
 - **Install**
 - **Block**
 - **Delete**
- **Key Contribution**

Literature Review

- **Existing Solutions**

- **VirusTotal**

- Aggregates 70+ antivirus engines.
- Provides detection ratios and technical reports.
- Reports can be difficult for beginners to understand.

- **Google Play Protect**

- Provides on-device Android application scanning.
- Provides minimal explanation to users.

MobSF :-

- Automated APK security analysis framework.
- Mainly designed for security researchers.
- Interface and analysis can be complex for ordinary users.

Existing System Limitations :-

- **Cryptic Reports** – Difficult for non-experts to understand.
- **Expert-Only UX** – Primarily designed for cybersecurity professionals.
- **Tool Fragmentation** – No unified platform for APKs, EXEs and URLs.
- **No AI Explanation** – Raw security results are not converted into actionable guidance.

Objectives

- 01. Static APK Analysis**

- Parse AndroidManifest.xml without executing the application.

- 02. Static EXE Analysis**

- Inspect PE headers, metadata and binaries.

- 03. SHA-256 Hashing**

- Generate hash fingerprints for verification.

04. Permission & Certificate Audit

- Detect dangerous permissions such as SMS and Camera.
- Validate developer signatures.

05. URL Analysis

- Check HTTPS, typosquatting and phishing indicators.

06. AI Explanation & Awareness

- Convert technical analysis results into simple language.

Primary Goal

Empower non-technical users to make informed and safe decisions before installation.

Required Tools and Technology

Component	Technology
Frontend	HTML5, CSS3, JavaScript (ES6)
Backend	Python 3.11
Framework	Flask
Database	SQLite
AI Engine	Gemini API
Libraries	hashlib, androguard, requests, os

Method/ Approach

- **Method / Approach**

- CyberInsight AI performs **pre-execution static analysis**, meaning uploaded files are analyzed without executing them.

- **Module 1 — APK Analysis**

- AndroidManifest analysis
- Permission analysis
- Certificate verification
- SHA-256 hash generation

- **Module 2 — EXE Analysis**

- PE metadata analysis
- Digital signature verification
- Suspicious string detection
- Embedded IPs, registry keys and APIs analysis

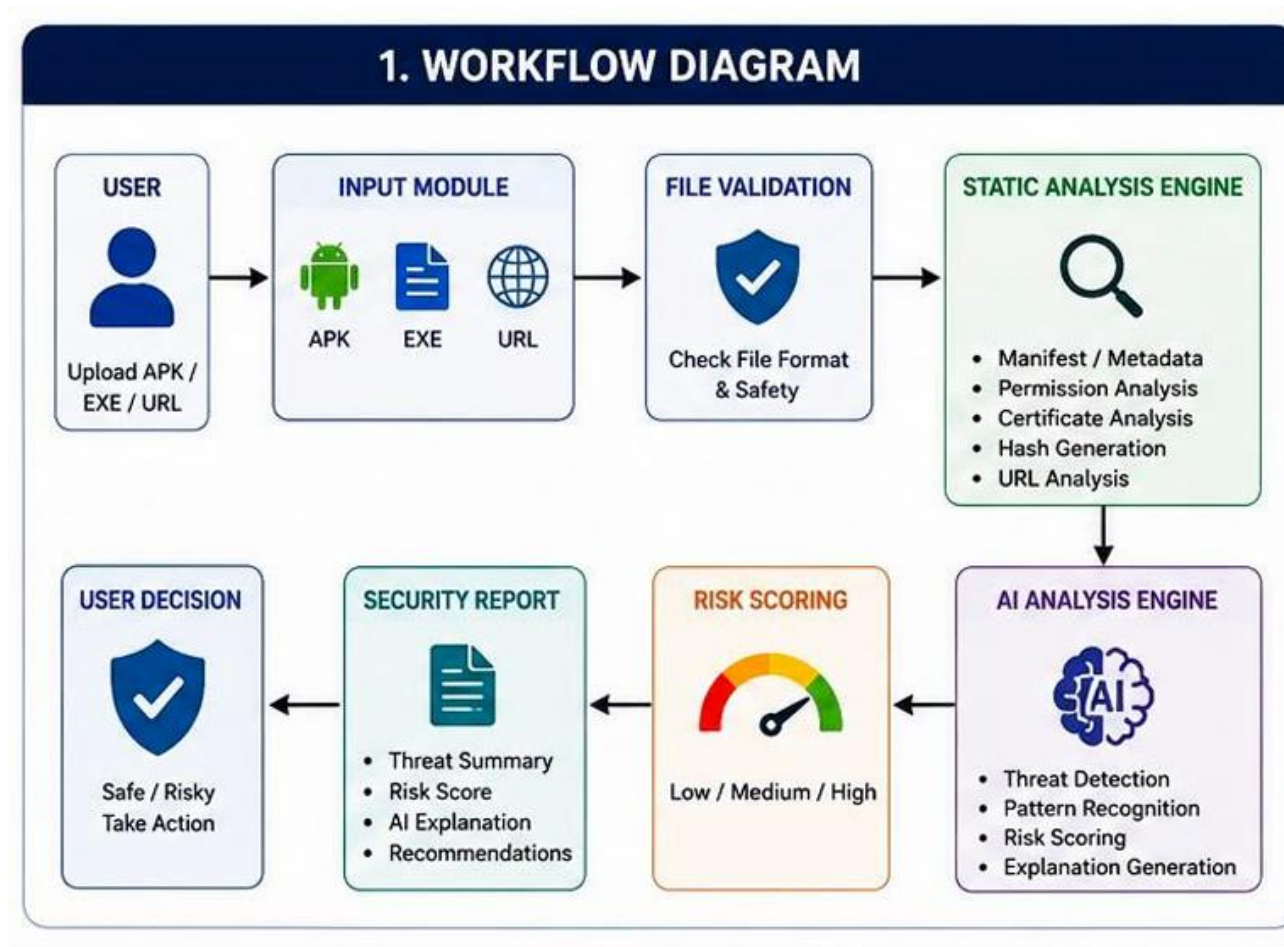
Module 3 — URL Analysis

- HTTPS verification
- Suspicious keyword detection
- Domain information analysis
- WHOIS and domain-age evaluation
- Typosquatting detection

Module 4 — AI Report

- Risk Score: **Low / Medium / High**
- Plain-language Threat Summary
- Recommendation:
 - **Install**
 - **Block**
 - **Delete**

SYSTEM WORKFLOW



COMPARISON

Feature	VirusTotal	CyberInsight AI
Analysis Engine	Multiple Antivirus Engines	Static Analysis
APK & EXE Analysis	Limited / Different Tools	Unified Analysis
Output	Technical Detection Logs	AI Plain-Language Explanation
Core Focus	Detection	Detection + Explanation
Target Users	Technical Users	Beginner Friendly
Actionability	Malware Counts	Risk Score + Recommendations

KEY PROJECT INNOVATIONS

Key Project Innovations

1. AI-Assisted Threat Explanation

- Converts binary signatures and forensic indicators into clear human-readable explanations.

2. Beginner-Friendly Interface

- Simple and intuitive interface designed for general computer users.

3. Unified Multi-Module Platform

- A single platform for:
 - APKs
 - EXEs
 - Web URLs

4. User Cyber Awareness

- Educates users about permission abuse and cybersecurity risks.
- Helps develop long-term security awareness.